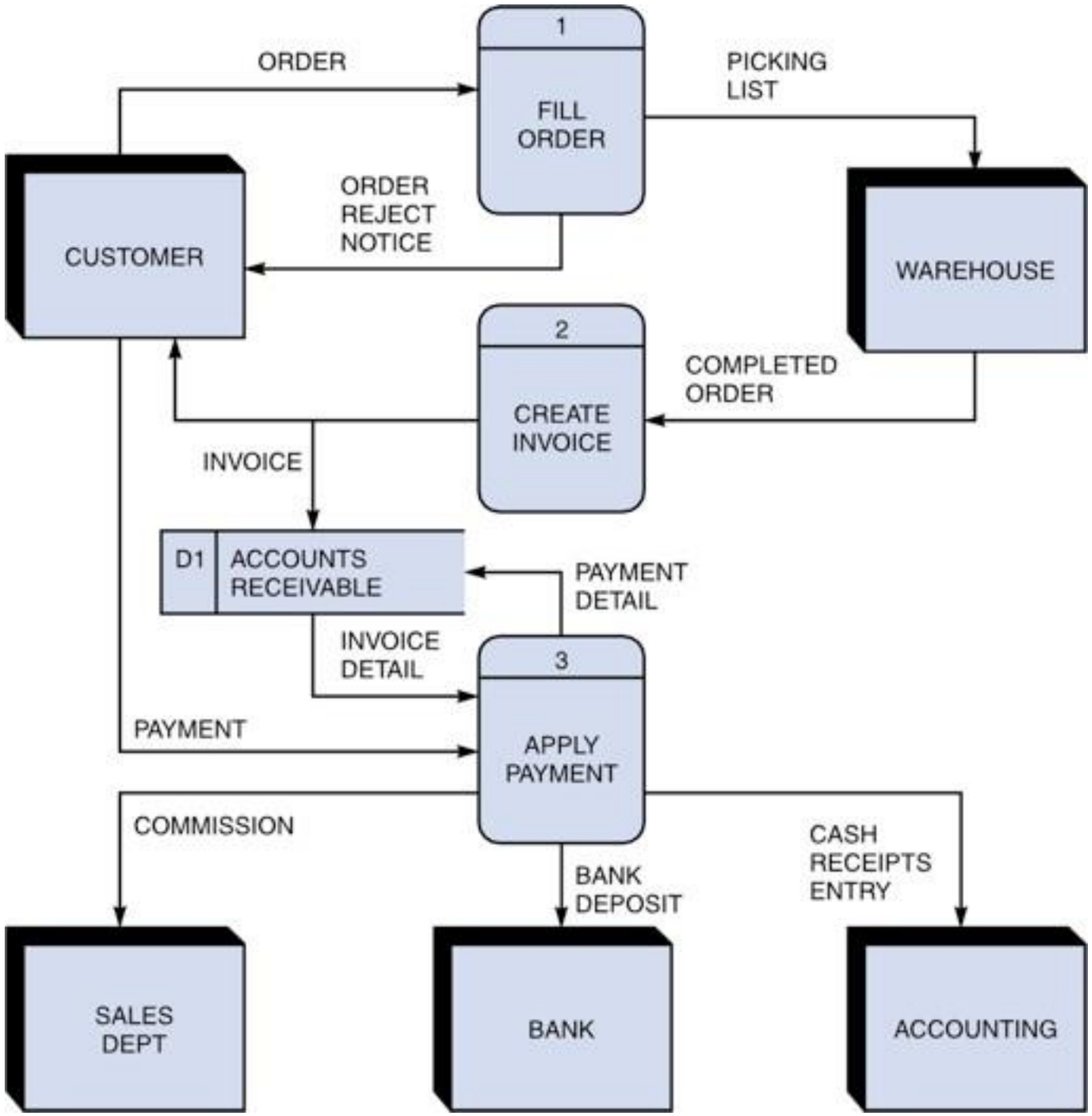
Data Flow Diagrams

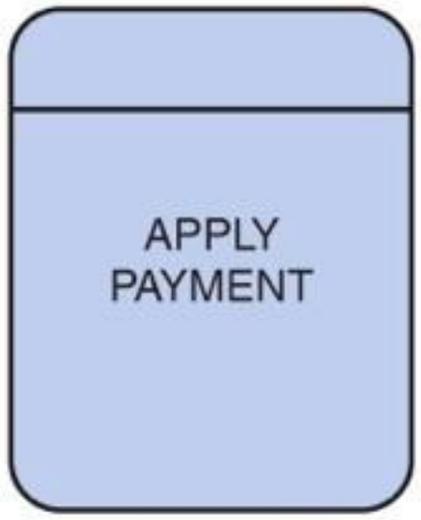
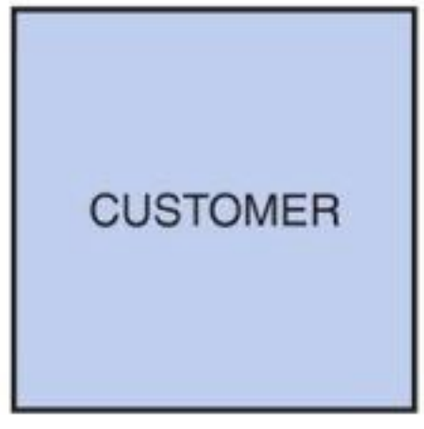
Chanduni Gamage
Faculty of Technology
University of Ruhuna

Acknowledgement
Essentials of Systems Analysis and Design (Valacich, George, Hoffer)
Systems analysis and Design (Shelly, Rosenblatt)

A data-flow diagram is a graphic that illustrates the movement of data between external entities and the processes and data stores within a system.

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Gane and Sarson Symbols	Symbol Name	Yourdon Symbols
	Process	
	Data Flow	
	Data Store	
	External Entity	

Terminologies ..

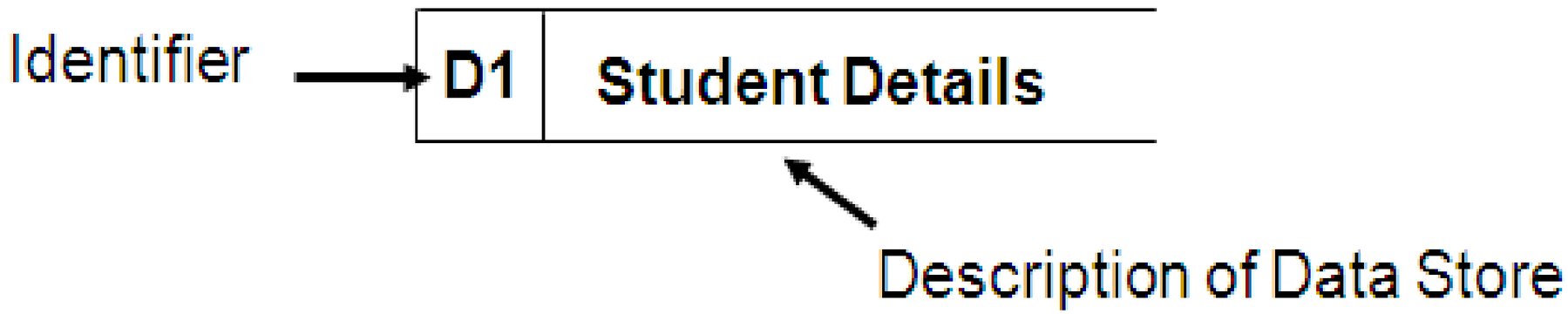
A data flow is a passage of information between objects.

A process is the work or actions performed on data so that they are transformed, stored, or distributed.

A source/sink is the origin and/or destination of the data. Source/sinks are sometimes referred to as external entities because they are outside the system.

Terminologies ..

A data store is data at rest. A data store might contain data about customers, students, customer orders, or supplier invoices.



- Every Data Store Identifier is unique –
- "M" denotes a Manual data store
- "D" denotes an Electronic data store
- "T" denotes a Transient data store

Process

A. No process can have only outputs. It is making data from nothing (a miracle). If an object has only outputs, then it must be a source.

B. No process can have only inputs (a black hole). If an object has only inputs, then it must be a sink.

C. A process has a verb-phrase label.

Data Store

D. Data cannot move directly from one data store to another data store. Data must be moved by a process.

E. Data cannot move directly from an outside source to a data store. Data must be moved by a process that receives data from the source and places the data into the data store.

F. Data cannot move directly to an outside sink from a data store. Data must be moved by a process.

G. A data store has a noun-phrase label.

Source/Sink

H. Data cannot move directly from a source to a sink. They must be moved by a process if the data are of any concern to our system. Otherwise, the data flow is not shown on the DFD.

I. A source/sink has a noun-phrase label.

Data Flow

J. A data flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The latter is usually indicated, however, by two separate arrows because the read and update usually happen at different times.

K. A fork in a data flow means that exactly the same data go from a common location to two or more different processes, data stores, or sources/sinks (it usually indicates different copies of the same data going to different locations).

L. A join in a data flow means that exactly the same data come from any of two or more different processes, data stores, or sources/sinks to a common location.

M. A data flow cannot go directly back to the same process it leaves. At least one other process must handle the data flow, produce some other data flow, and return the original data flow to the beginning process.

N. A data flow to a data store means update (delete or change).

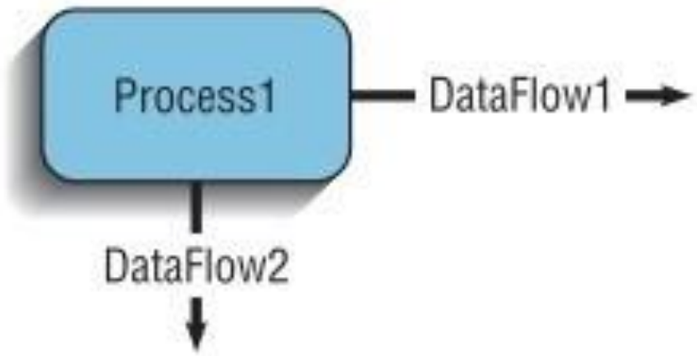
O. A data flow from a data store means retrieve or use.

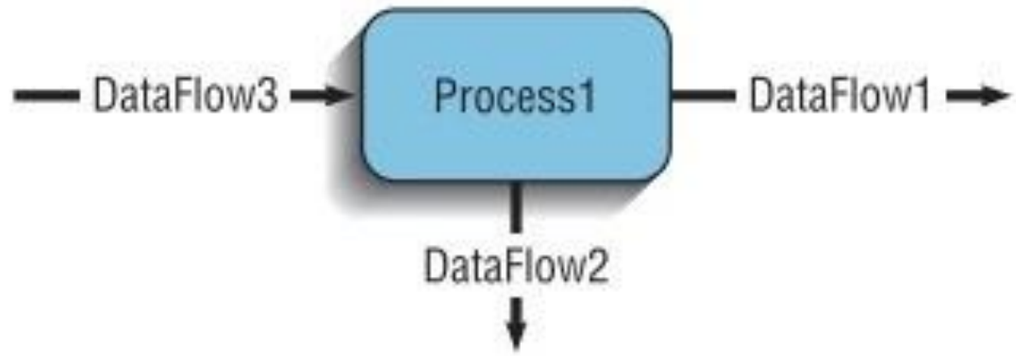
P. A data flow has a noun-phrase label. More than one data-flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.

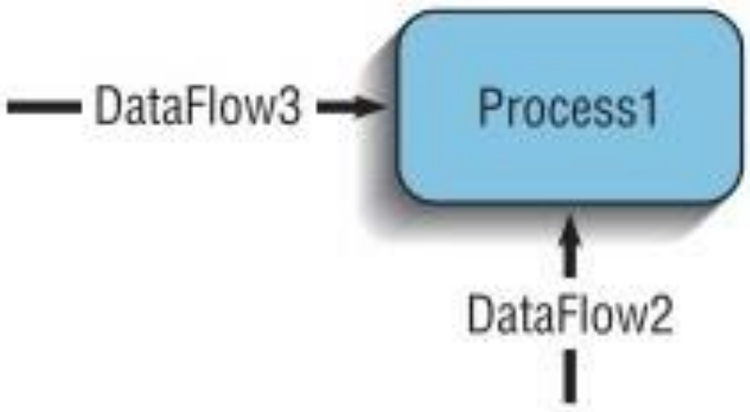
Source: Based on J. Celko, "I. Data Flow Diagrams," *Computer Language* 4 (January 1987), 41–43.

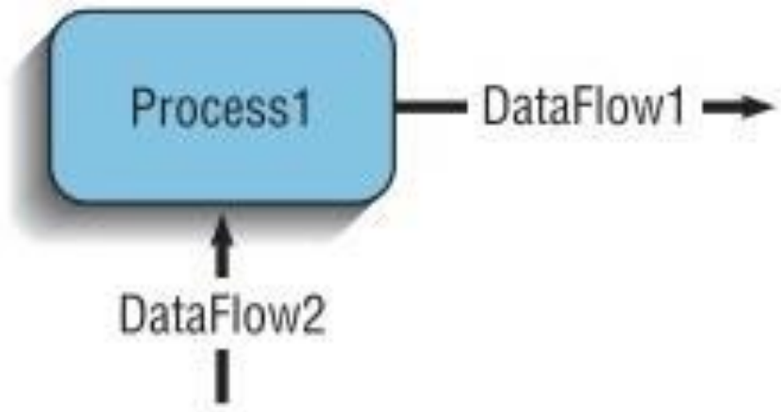
In summary ..

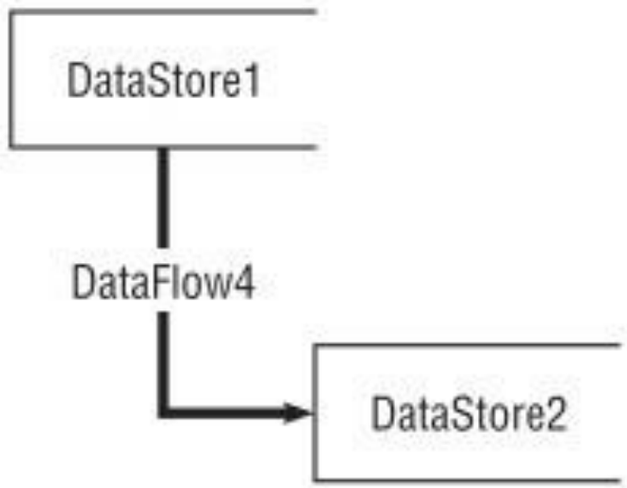
Process	Data store	External entity	Data flow
No process can have only outputs	Data cannot directly move from one data store to another data store	Data cannot directly move from asource to a sink	
No process can have only inputs	Data cannot directly move from one data store to an external entity		

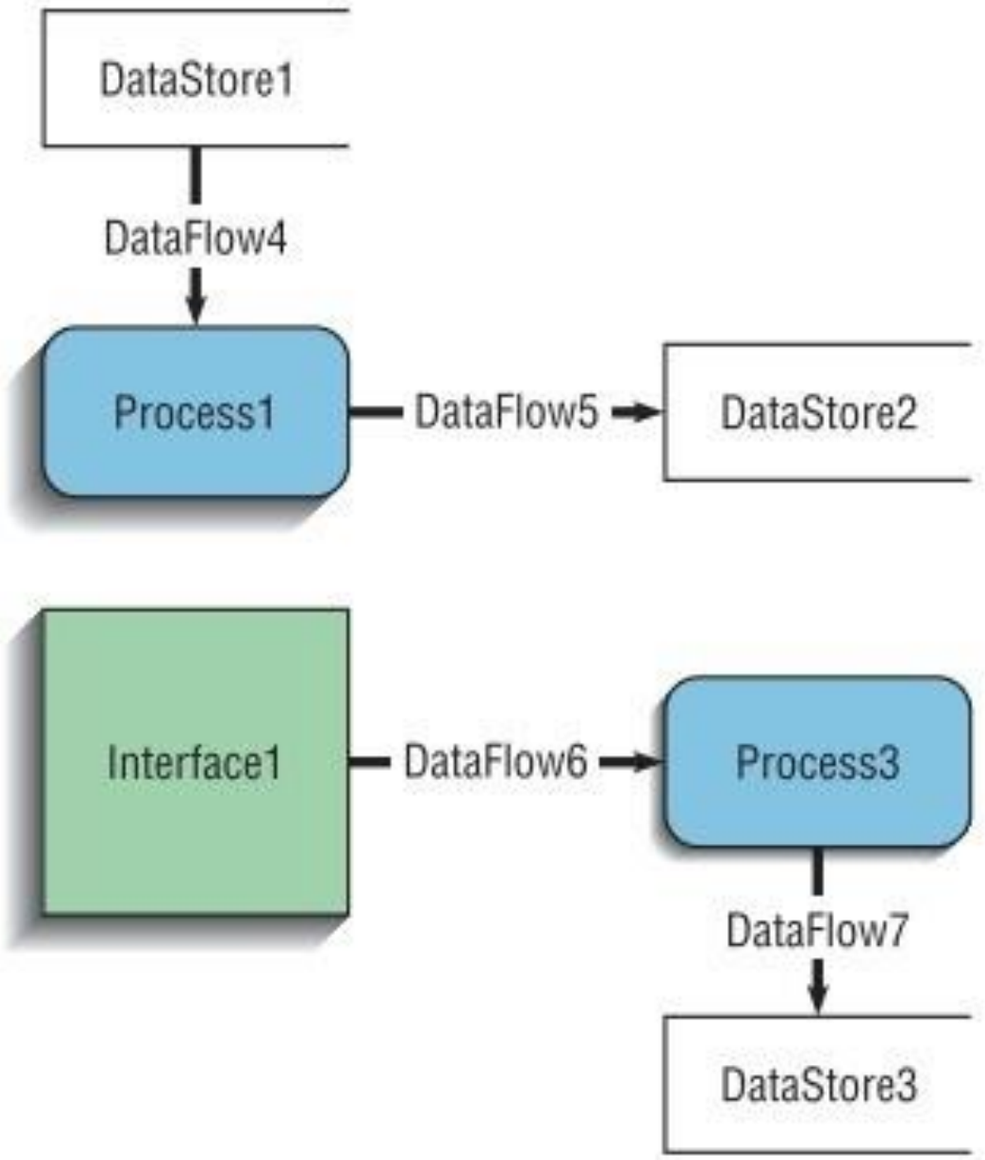

Process1
DataFlow1
DataFlow2

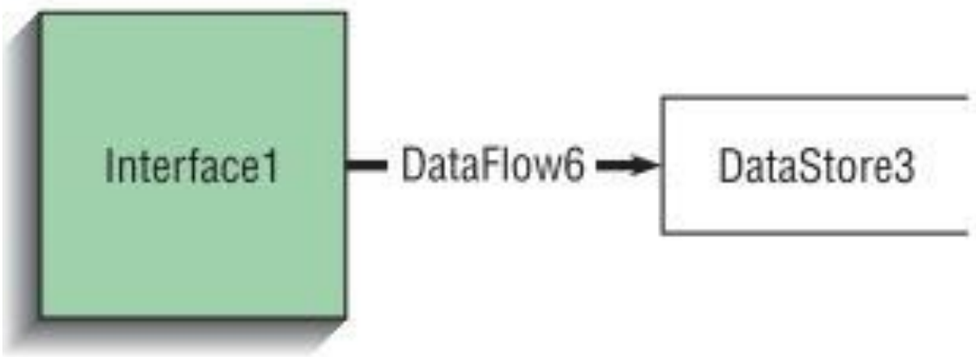

DataFlow3
Process1
DataFlow1
DataFlow2

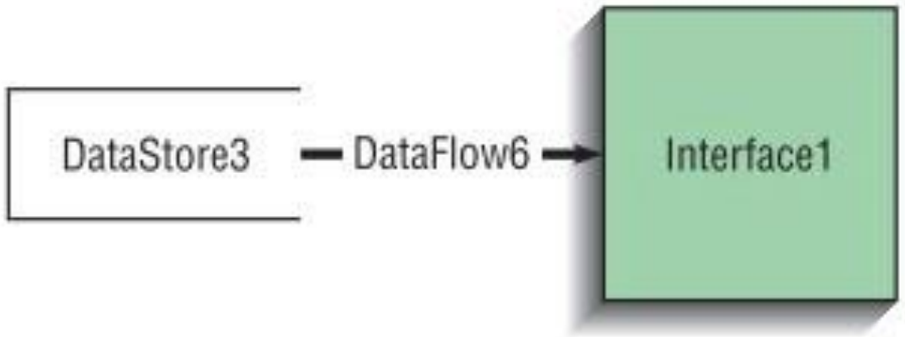

DataFlow3
Process1
DataFlow2

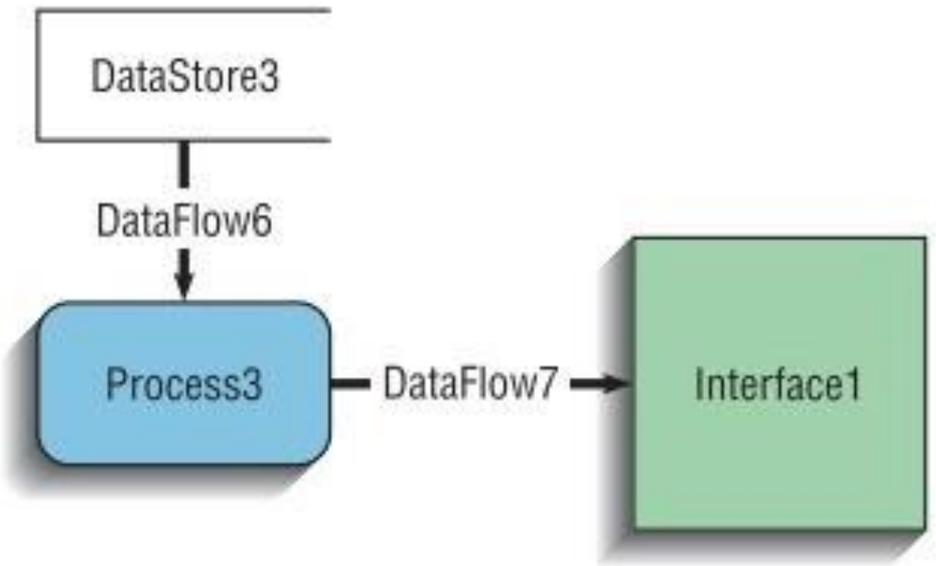

Process1
DataFlow1
DataFlow2

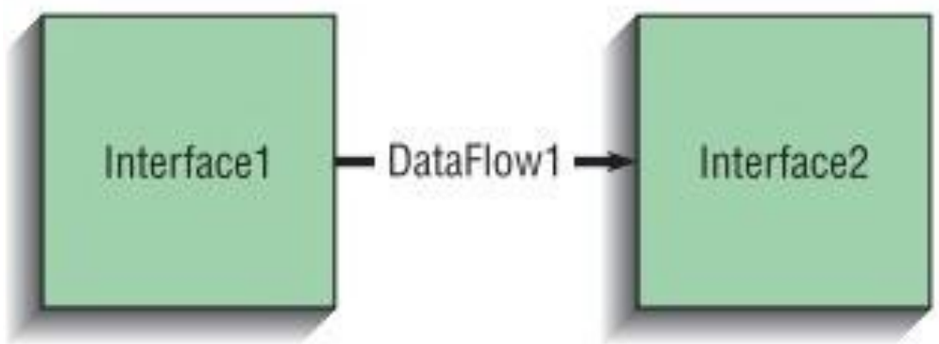

DataStore1
DataFlow4
DataStore2

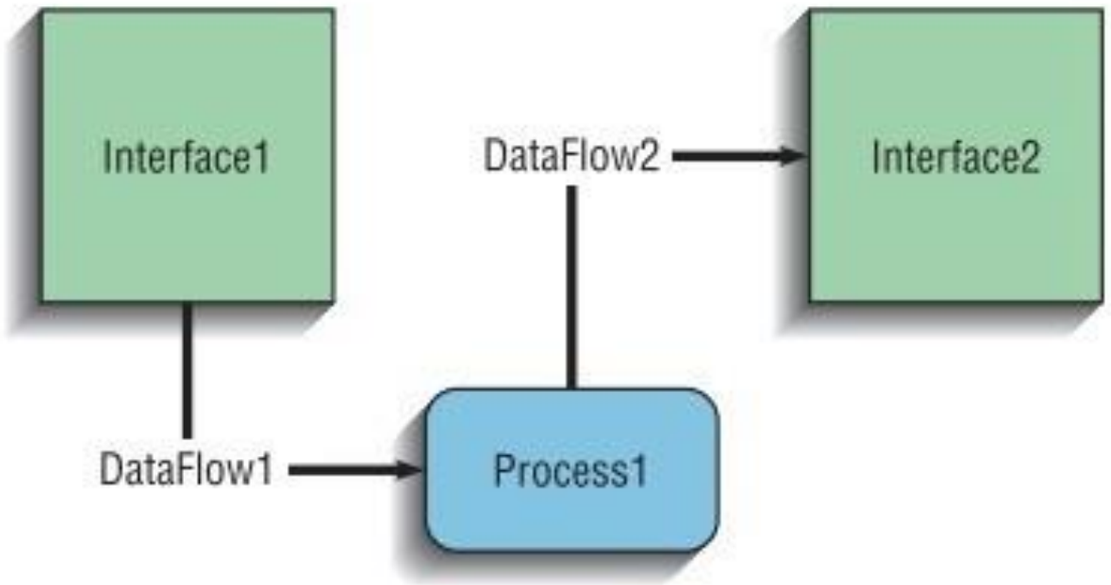

DataStore1
DataFlow4
Process1
DataFlow5
DataStore2
Interface1
DataFlow6
Process3
DataFlow7
DataStore3

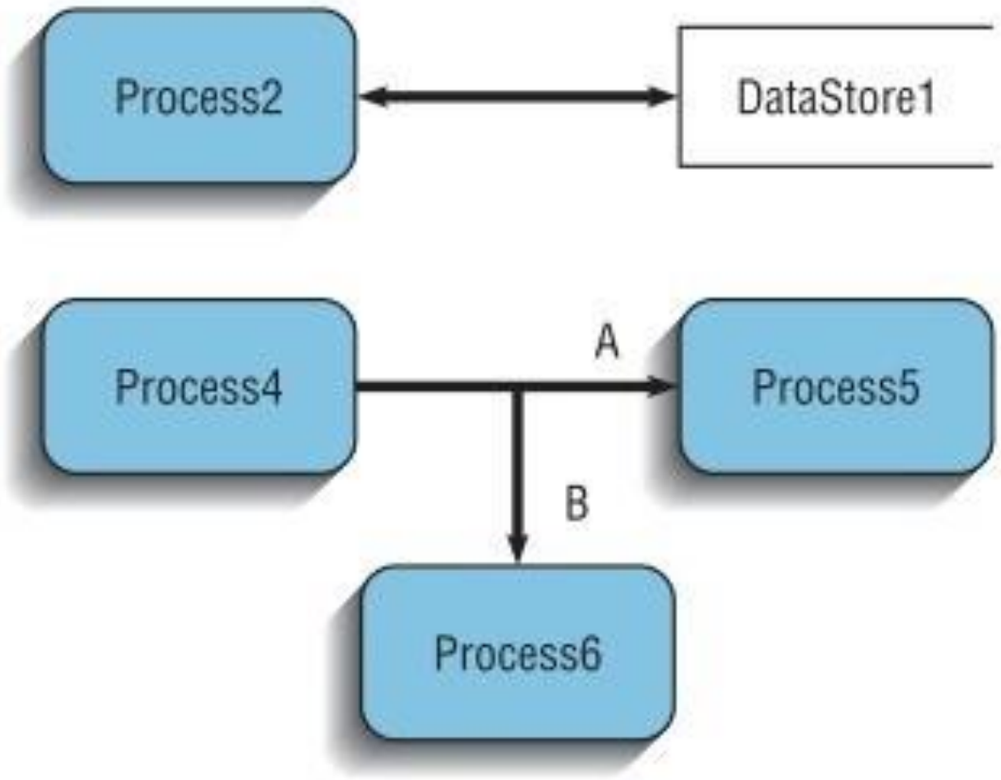

Interface1
DataFlow6
DataStore3

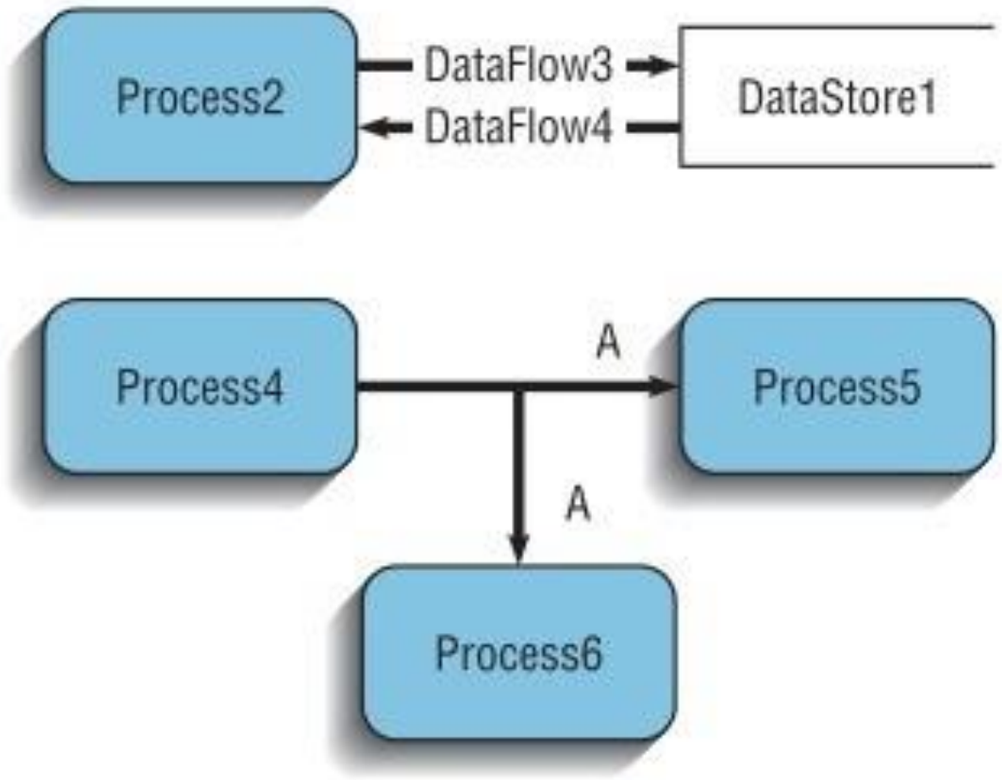

DataStore3
DataFlow6
Interface1

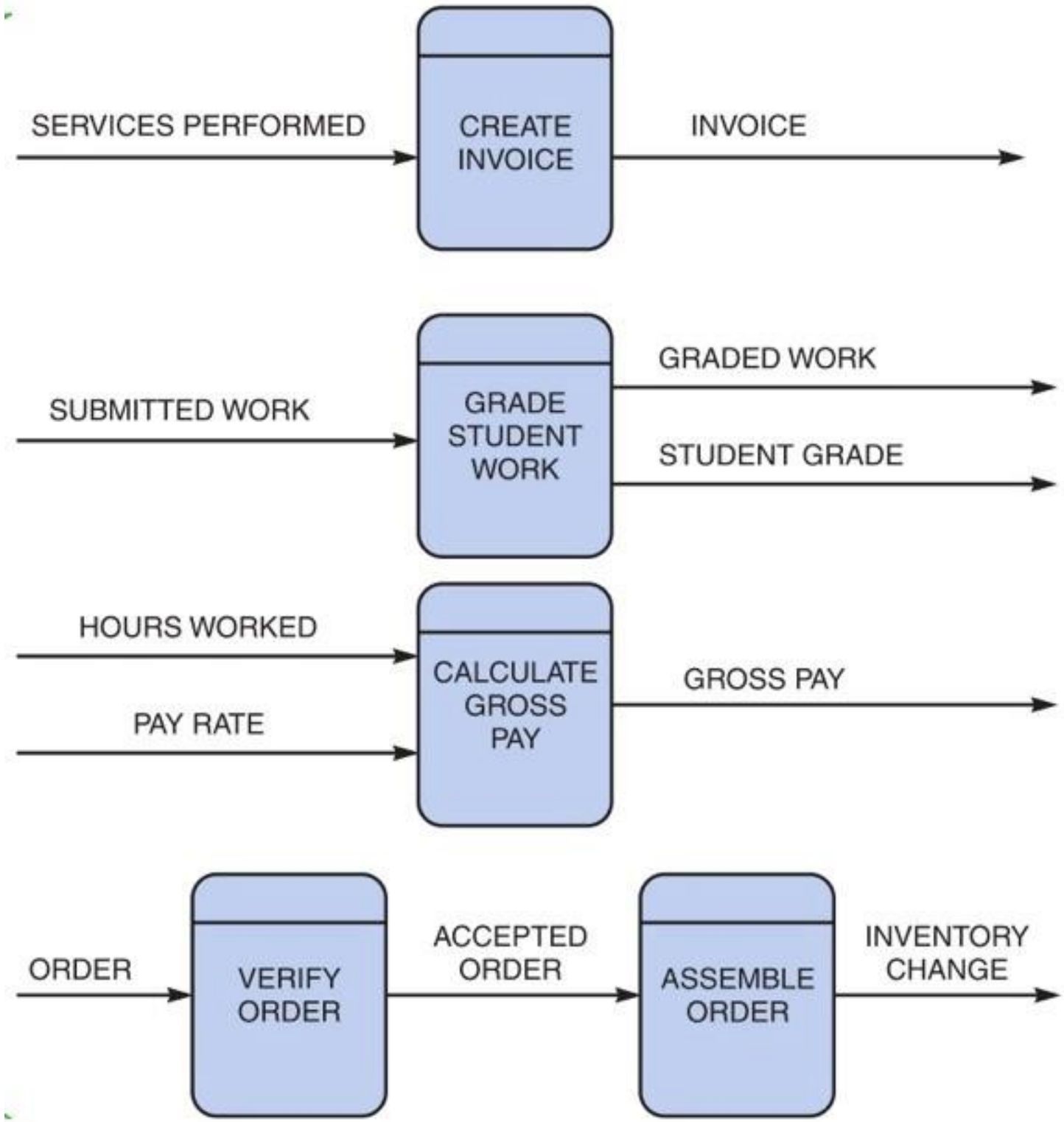

DataStore3
DataFlow6
Process3
DataFlow7
Interface1

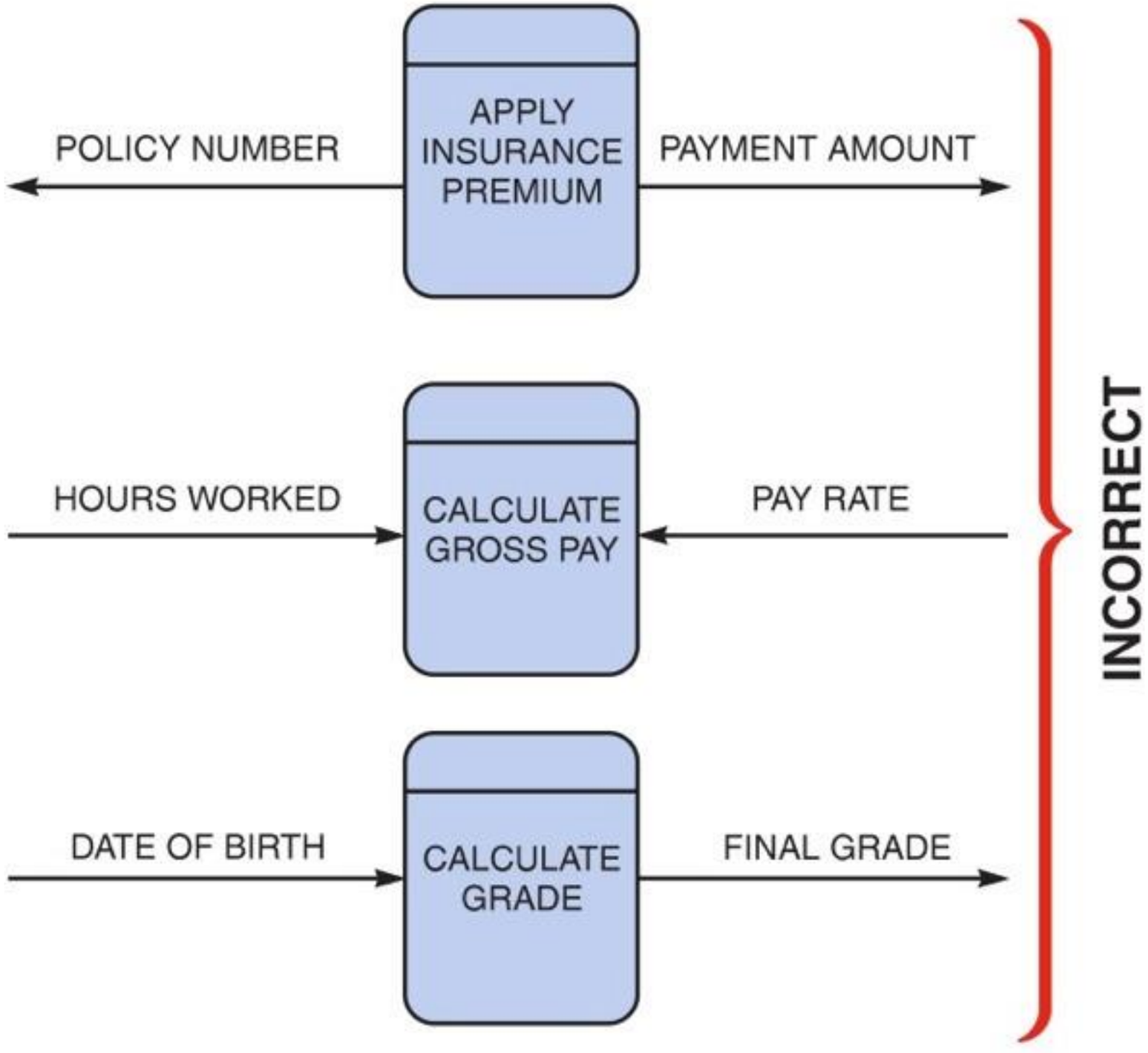

Interface1
DataFlow1
Interface2


Interface1
DataFlow2
Interface2
DataFlow1
Process1


Process2
DataStore1
Process4
A
Process5
B
Process6


Process2
DataFlow3
DataFlow4
DataStore1
Process4
A
Process5
A
Process6


SERVICES PERFORMED
CREATE INVOICE
INVOICE
SUBMITTED WORK
GRADE STUDENT WORK
GRADED WORK
STUDENT GRADE
HOURS WORKED
PAY RATE
CALCULATE GROSS PAY
GROSS PAY
ORDER
VERIFY ORDER
ACCEPTED ORDER
ASSEMBLE ORDER
INVENTORY CHANGE


APPLY INSURANCE PREMIUM
POLICY NUMBER
PAYMENT AMOUNT
CALCULATE GROSS PAY
HOURS WORKED
PAY RATE
CALCULATE GRADE
DATE OF BIRTH
FINAL GRADE
INCORRECT

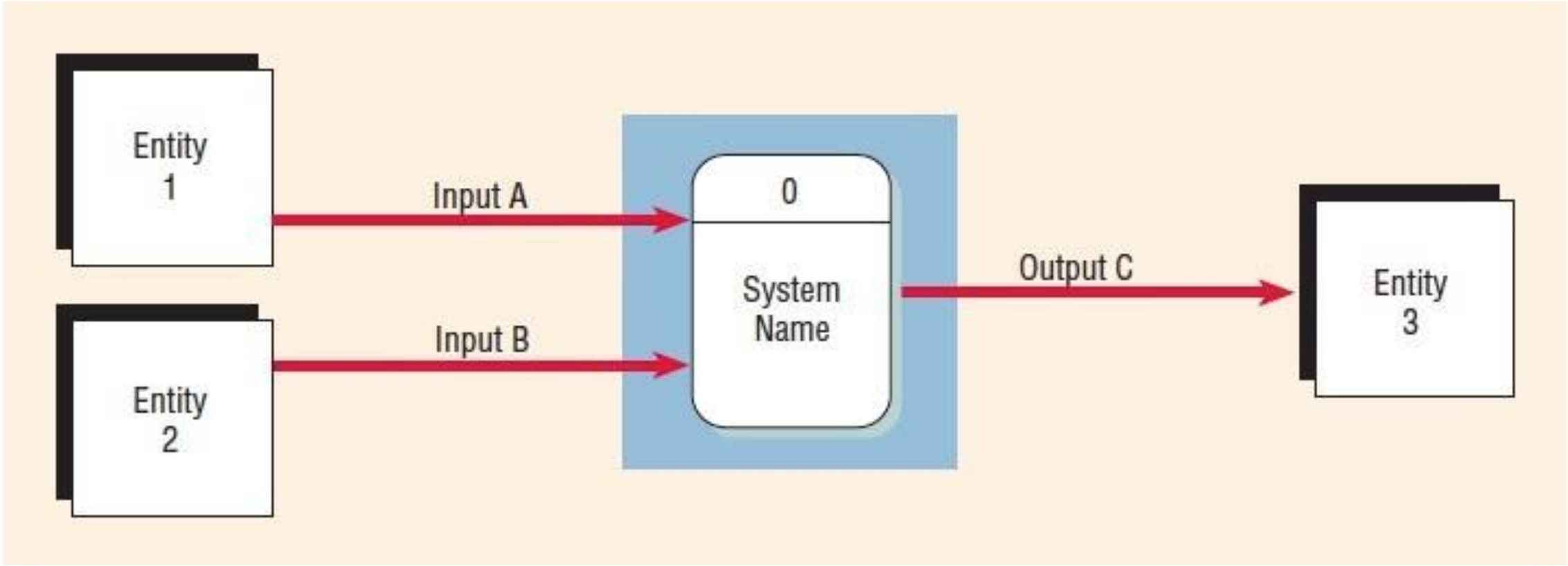
The first step in constructing aset of DFDs is to draw a context diagram. A context diagram is a top-level view of an information system that shows the system's boundaries and scope.

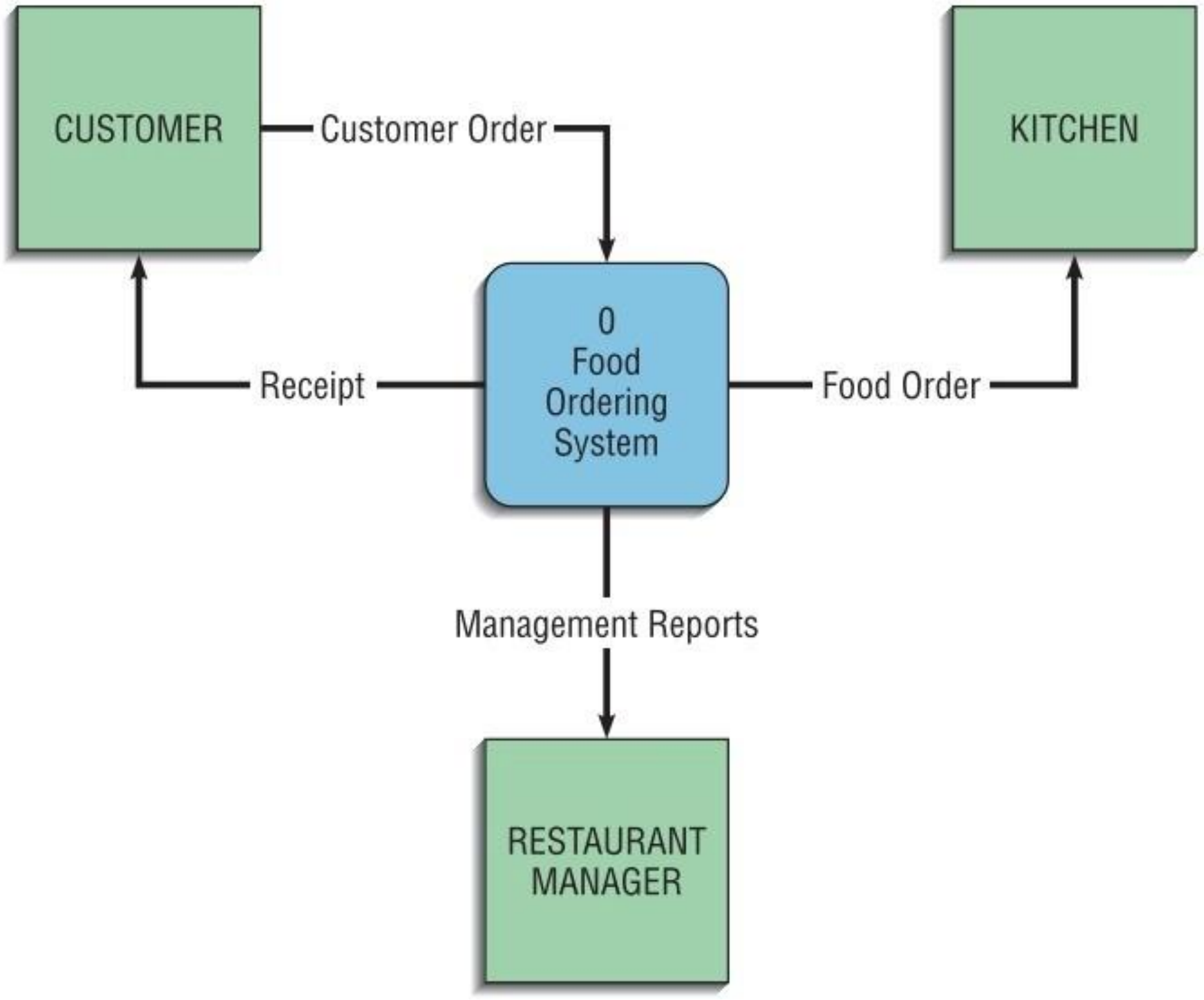
To draw acontext diagram, you start by placing asingle process symbol in the center of the page. The symbol represents the entire information system, and you identify it asprocess 0 (the numeral zero, and not the letter O). Then you place the system entities around the perimeter of the page and use data flows to connect the entities to the central process.

Data stores are not shown in the context diagram because they are contained within the system and remain hidden until more detailed diagrams are created.

Context Diagram

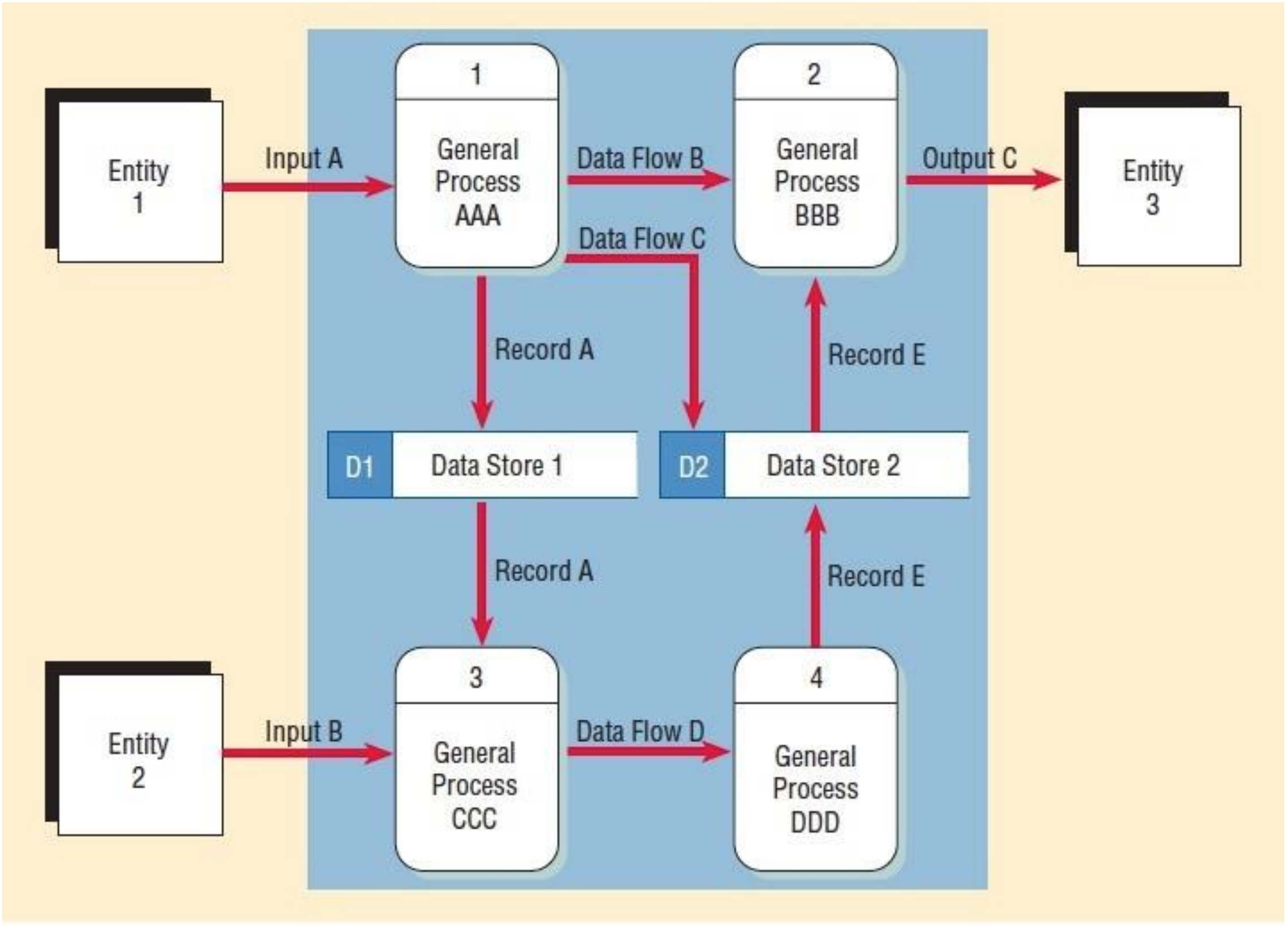
- A level 0 DFD, is also known as a fundamental system model or a context model.
- It represents the entire software element as a single process with input and output data.
- All external entities should be identified and shown.
- Rules:
  - ✓ Only one process.
  - ✓ Data flows should be labeled.
  - ✓ No data store can be shown in context diagrams as they are internal, to the systems only.

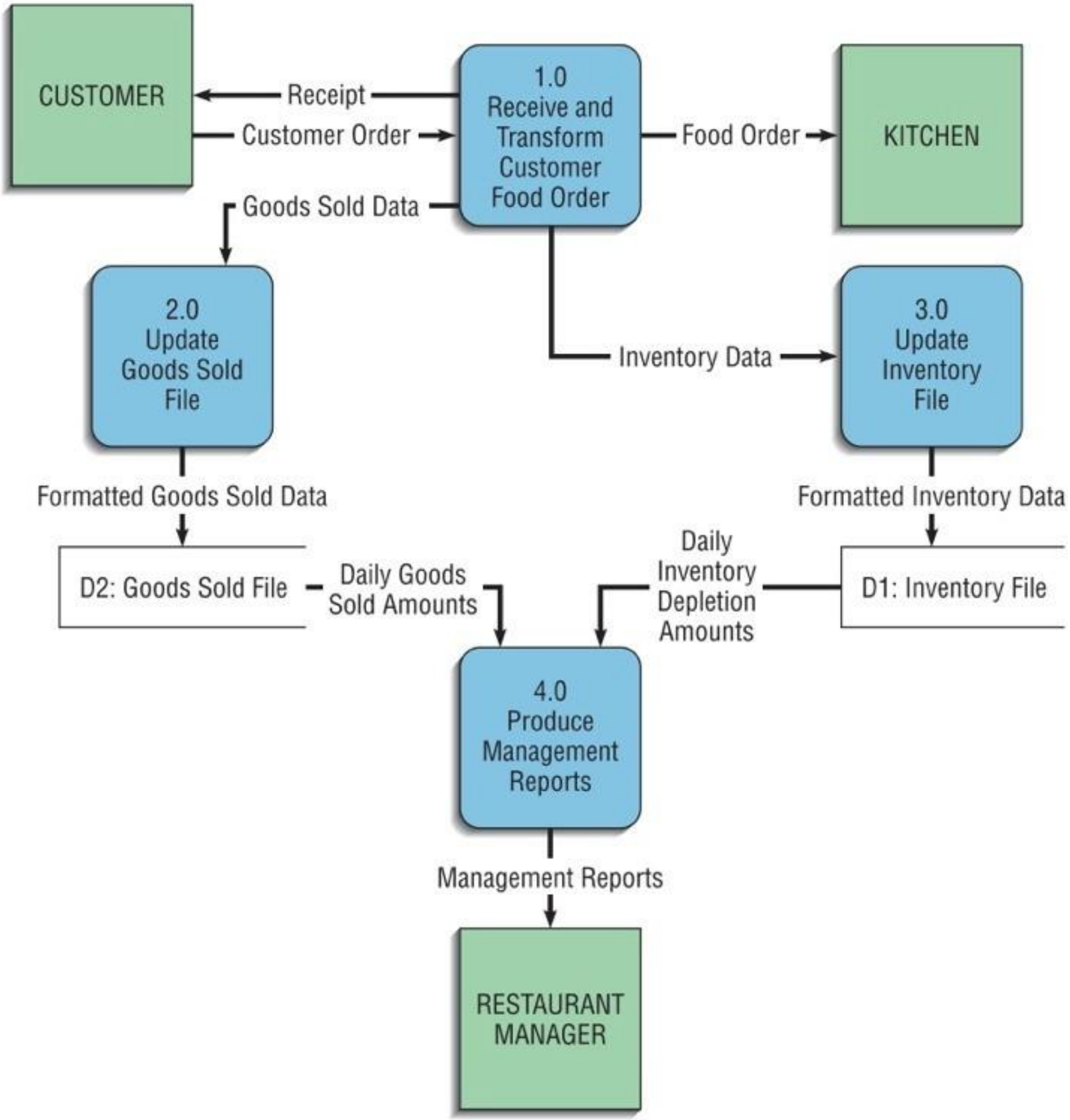

Entity
1
Entity
2
Input A
Input B
0
System
Name
Output C
Entity
3


CUSTOMER
Customer Order
KITCHEN
0
Food
Ordering
System
Receipt
Food Order
Management Reports
RESTAURANT
MANAGER

Level 1 Data Flow Diagram

- It's a more refined version of the context diagram.
- It covers the entire system; all the main processes are shown.
- The DFD should be balanced with respect to the context diagram
  - ✓ No new external entities should be there.
  - ✓ The data flows from the context diagrams should be visible.
- Rules:
  - ✓ Repetition of data sources is allowed.
  - ✓ Process cannot be repeated


Entity
1
Input A
1
General
Process
AAA
Data Flow B
2
General
Process
BBB
Output C
Entity
3
Data Flow C
Record A
Record E
D1
Data Store 1
D2
Data Store 2
Record A
Record E
Entity
2
Input B
3
General
Process
CCC
Data Flow D
4
General
Process
DDD


CUSTOMER
Receipt
Customer Order
Goods Sold Data
1.0
Receive and
Transform
Customer
Food Order
Food Order
KITCHEN
2.0
Update
Goods Sold
File
Inventory Data
3.0
Update
Inventory
File
Formatted Goods Sold Data
Formatted Inventory Data
D2: Goods Sold File
Daily Goods
Sold Amounts
Daily
Inventory
Depletion
Amounts
D1: Inventory File
4.0
Produce
Management
Reports
Management Reports
RESTAURANT
MANAGER

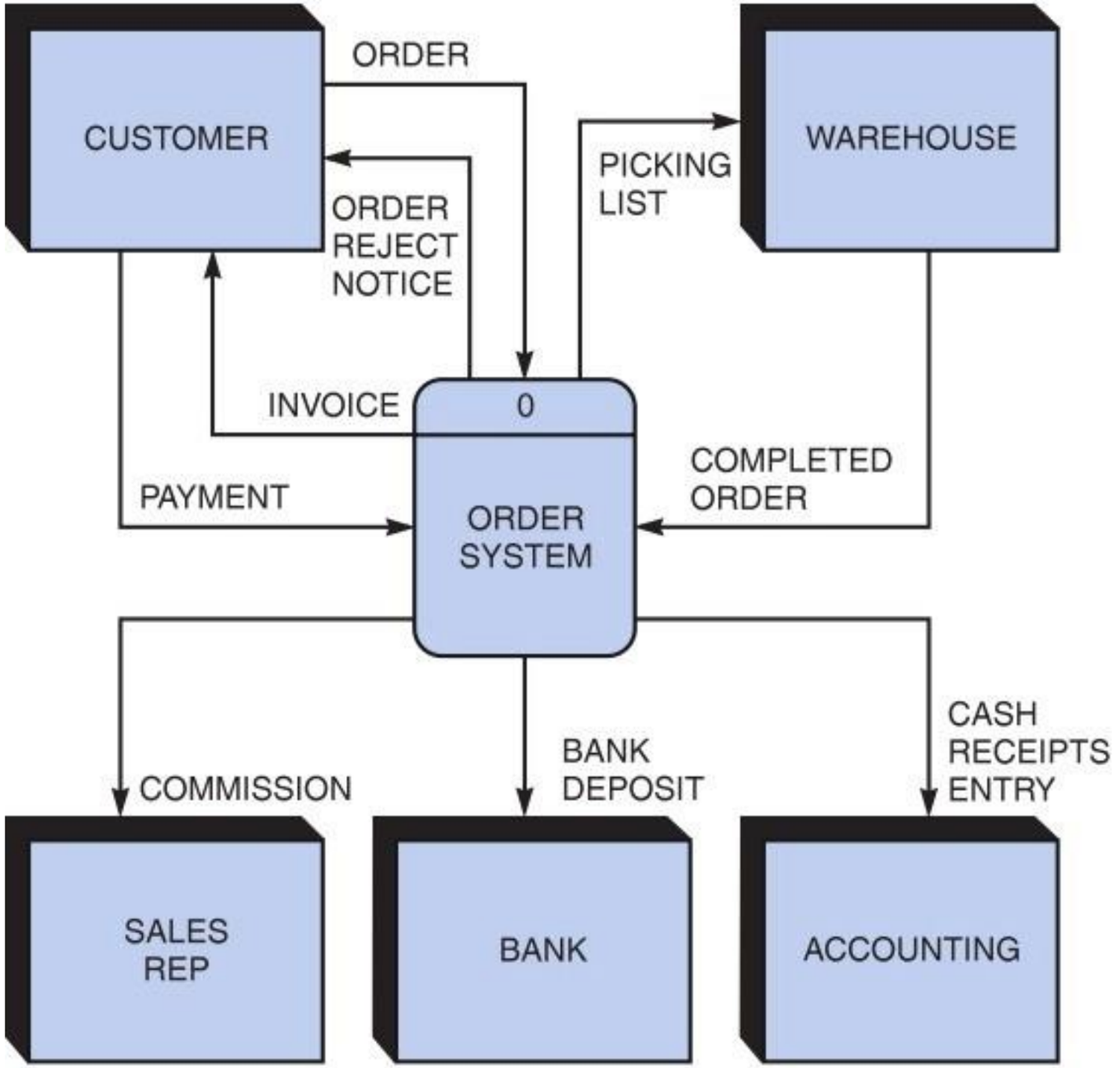
EXAMPLE: LEVEL 0 DIAGRAM FOR THE ORDER SYSTEM

1. A CUSTOMER submits an ORDER. Depending on the processing logic, the FILL ORDER process either sends an ORDER REJECT NOTICE back to the customer or sends a PICKING LIST to the WAREHOUSE.

2. A COMPLETED ORDER from the WAREHOUSE is input to the CREATE INVOICE process, which outputs an INVOICE to both the CUSTOMER process and the ACCOUNTS RECEIVABLE data store.

3. A CUSTOMER makes a PAYMENT that is processed by APPLY PAYMENT. APPLY PAYMENT requires INVOICE DETAIL input from the ACCOUNTS RECEIVABLE data store along with the PAYMENT. APPLY PAYMENT also outputs PAYMENT DETAIL back to the ACCOUNTS RECEIVABLE data store and outputs COMMISSION to the SALES REP, BANK DEPOSIT to the BANK, and CASH RECEIPTS ENTRY to ACCOUNTING.

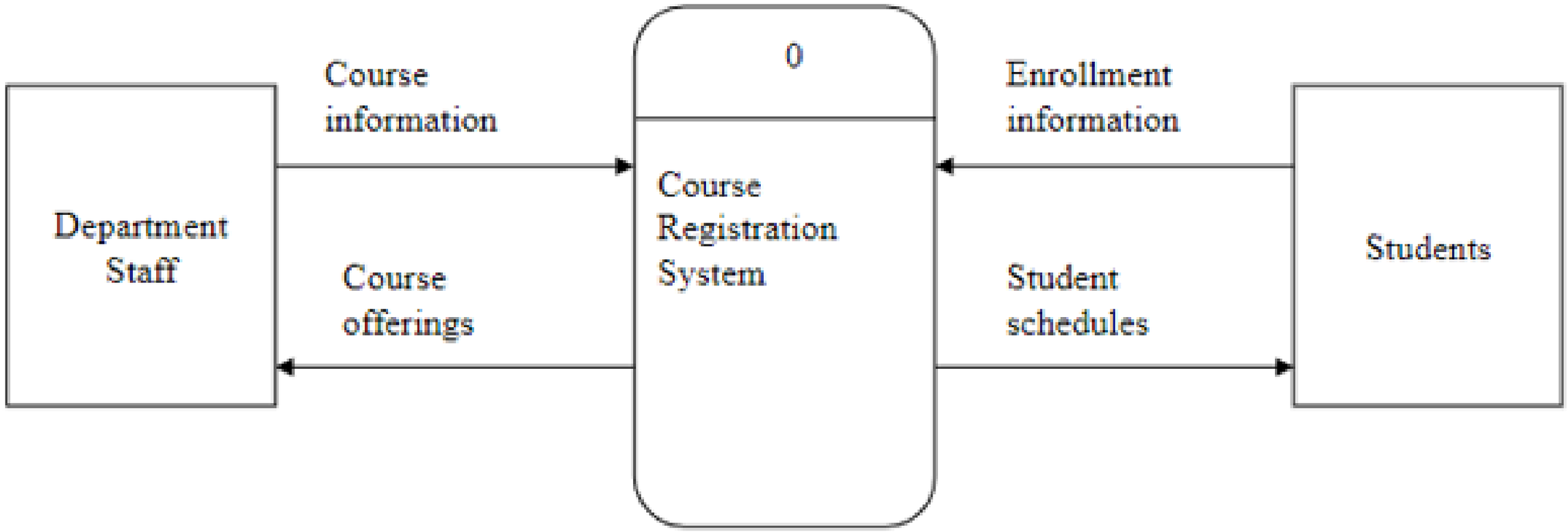
EXAMPLE: CONTEXT DIAGRAM FOR AN ORDER SYSTEM



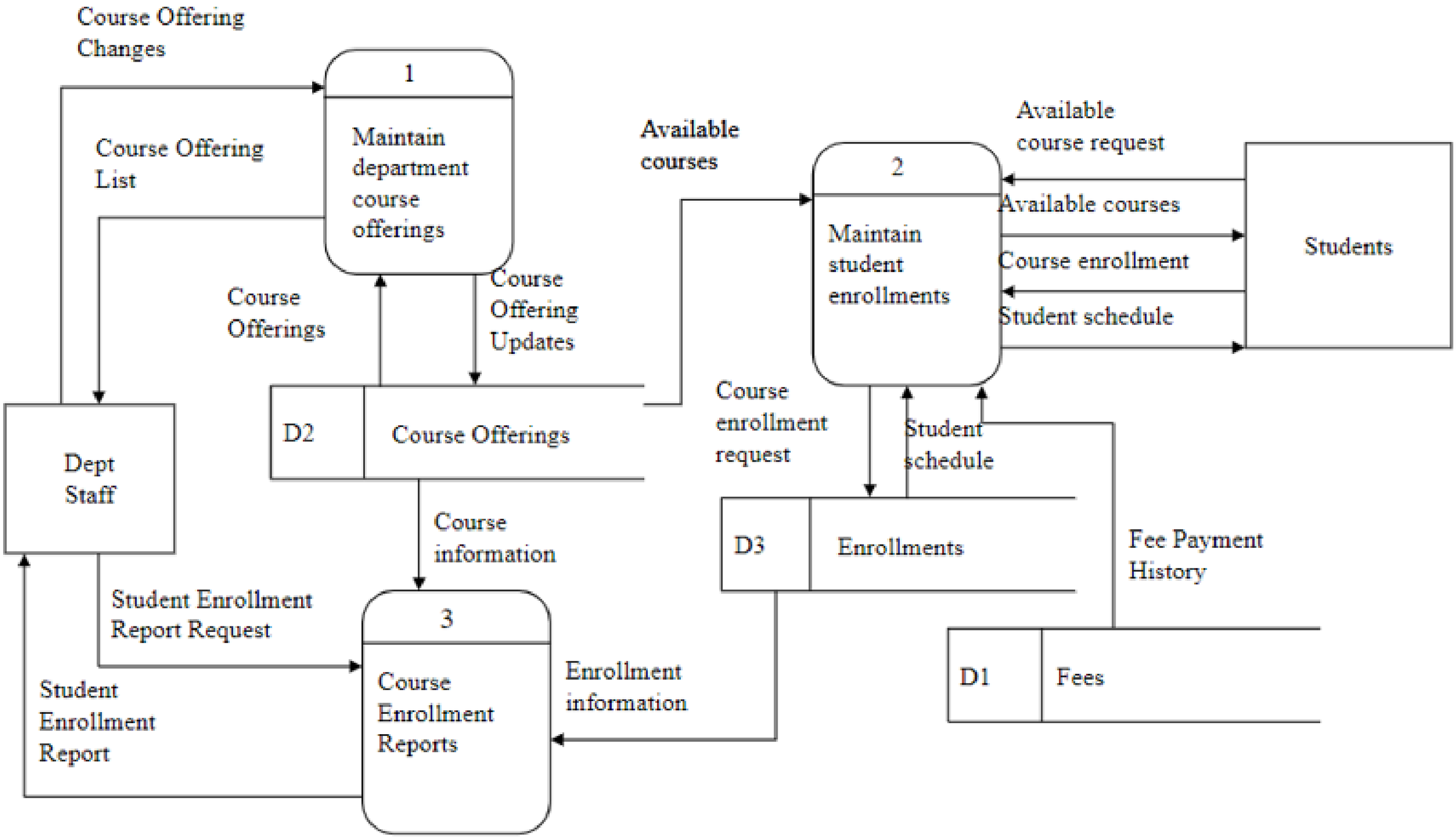
EXAMPLE: COURSE REGISTRATION SYSTEM

The online course registration system should enable the staff of each academic department to examine the course offered by their department, add and remove courses, and change the information about them (E.g: the maximum number of students). It should permit students to examine their schedules and examine the course in which they are enrolled. Department staff should be able to print a variety of reports about the courses and the students enrolled in them. They should ensure that no student takes too many courses and that the students who have any unpaid fees are not permitted to register. (Assume that a fees data store is maintained by the university's financial office that the registration system accesses but doesn't change).

EXAMPLE: LEVEL 0 DFD



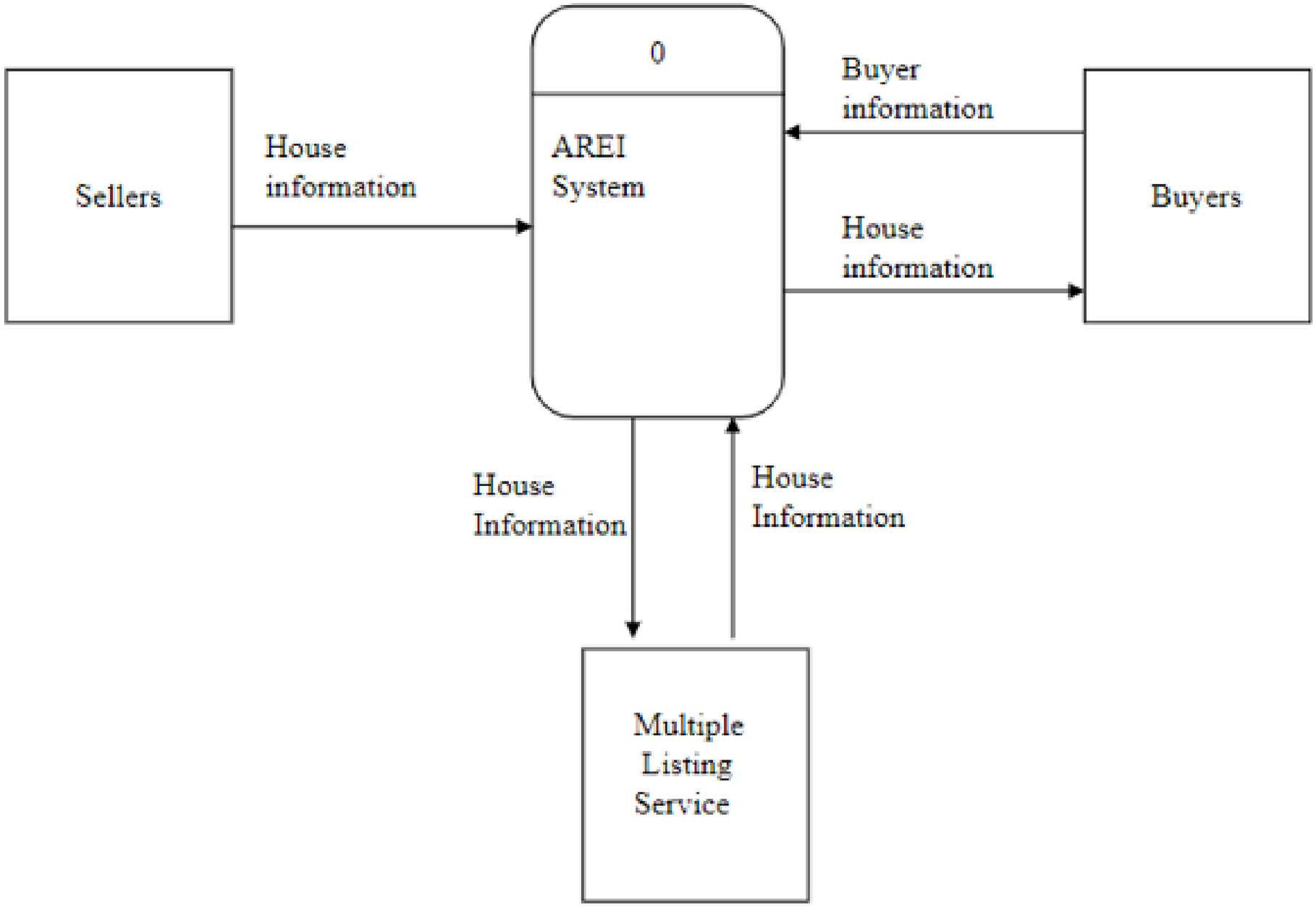
EXAMPLE: LEVEL 1 DFD



EXERCISE: AREI SYSTEM

A Real Estate Inc. (AREI) sells houses. People who want to sell their houses sign a contract with AREI and provide information on their house. This information is kept in a database by AREI, and a subset of this information is sent to the citywide multiple listing service used by all real estate agents. AREI works with two types of potential buyers. Some buyers have an interest in one specific house. In this case, AREI prints information from its database, which the real estate agent uses to help show the house to the buyer (a process beyond the scope of the system to be modeled). Other buyers seek AREI's advice in finding a house that meets their needs. In this case, the buyer completes a buyer information form that is entered into a buyer database, and AREI real estate agents use its information to search AREI's database and the multiple-listing service for houses that meet their needs. The results of these searches are printed and used to help the real estate agent show houses to the buyer.

EXAMPLE: LEVEL 0 DFD



EXAMPLE: LEVEL 1 DFD

